

# Gabrielle Butler

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## SUMMARY

Recent Biology graduate with a minor in Bioinformatics and hands-on experience in both wet and dry lab techniques. Skilled in DNA extraction, PCR, gel electrophoresis, bacterial and yeast culture, microscopy, aseptic technique, and standard laboratory instrumentation. Experienced in Python-based genomic data analysis, querying bioinformatics databases, and automating data workflows. Coauthor on a manuscript analyzing SNPs in CYP3A regulatory regions for pharmacogenomics. Trained in lab safety, hazardous waste management, and animal biotechnology. Bilingual in English and Spanish.

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## EDUCATION

### University of Florida

August 2025

B.A., Major in Biology, Minor in Bioinformatics

Gainesville, FL

- **Relevant Coursework:** Biology I & II with labs, General Chemistry I & II with labs, Organic Chemistry I, Eukaryotic Cell Structure & Function, Genetics, Basic Biology of Microorganisms with lab, Genomics and Biotechnology, Anatomy & Physiology with lab, Cell Systems and Physiology, Intro to Bioinformatics, Python Programming for Biologists, Bioinformatics Research, Critical Analysis of Biological Research

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## RESEARCH EXPERIENCE

### University of Florida - College of Medicine - Department of Pharmacotherapy and Translational Research,

#### Dr. Danxin Wang Lab

Oct. 2024 – Aug. 2025

Research Assistant

Gainesville, FL

- Designed and implemented a **bioinformatics pipeline** to analyze SNP interactions within transcription factor binding sites (TFBS) in regulatory regions of CYP3A pharmacogenes, contributing to pharmacogenomics and personalized medicine research.
- Queried and integrated data from NCBI, Ensembl, UCSC Genome Browser, UniProt, GTEx, KEGG, PDB, UniBind, BRENDA, gnomAD, SNP Nexus, and the Human TF Database.
- Used **Python (requests, automation scripts)** to access APIs and streamline data collection and analysis.
- Applied computational methods to regulatory region analysis using 4C, histone modification markers (H3K4me1, H3K27ac), ATAC-Seq, and GRO-Seq datasets.
  - Identified 3 SNPs with potential clinical significance in drug metabolism.
  - Coauthor on a research paper (expected publication Winter 2025).

### University of Florida - Genetics Institute, Dr. Eric Wang Lab

Aug. 2023 – Feb.

2025

Laboratory Assistant

Gainesville, FL

- Conducted **DNA Extraction, Polymerase Chain Reactions (PCRs), and Agarose Gel Electrophoresis and Imaging** on mouse DNA
- Operated thermocyclers, centrifuges, micropipettes, vortex mixers, and gel imaging systems.
- Maintained a lab bench and a lab notebook

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## SKILLS and CERTIFICATIONS

- **Bioinformatics & Programming:** Python (data analysis, API querying, automation), R (introductory), SQL (introductory), PowerShell, Excel, Access
- **Databases & Tools:** NCBI, Ensembl, UCSC Genome Browser, UniProt, UniBind, GTEx, KEGG, PDB, BRENDA, gnomAD, SNP Nexus, Human TF Database
- **Sequence Analysis:** BLAST, Clustal Omega, FASTA/FASTQ, Multiple Sequence Alignment, Primer Design
- **Wet Lab Techniques:** DNA extraction and purification, PCR (standard), Gel electrophoresis (agarose) and imaging, DNA quantification, Bacterial and yeast cell culture, Aseptic/sterile technique, Gram staining and

- microscopy (compound light), Media and buffer preparation, reagent handling, laboratory safety and biosafety
- **Instrumentation:** Pipette and pipette aspirators, PCR thermal cyclers, Electrophoresis and gel documentation systems (UV), Centrifuges: microcentrifuge, high-speed, Heat blocks and hot plates, -20 °C freezers, Fume hood and biosafety cabinets (BSL-1/2), Compound light microscope
- **Certifications:**
  - University of Florida Lab Training: Hazardous and Biomedical Waste Management, Bloodborne Pathogens, HIPAA Awareness, Animal Care Services, Mouse and Rat Biomethodology
  - Passed the Biology (6-12) Florida Subject Area Test for the Temporary Teaching Certificate
- **Languages:** Spanish (Professional), English (Native)

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## WORK EXPERIENCE

### Wawa

May 2023 – Aug. 2023

Associate

Ormond Beach, FL

- Gained transferable skills in teamwork, communication, and problem-solving in high-paced environments.
- Worked register, deli, and beverage area
- Developed customer service skills

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### Publix

Feb. 2022 – Aug. 2022

Deli Clerk

Ormond Beach, FL

- Sliced meats, prepared and assembled healthy foods and subs, created a friendly work environment, regularly engaged with customers, handled hot food items, responsible for closing the store (cleaned deli, prepared for the next day)

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### Tropical Smoothie Café

Apr. 2021 – Aug. 2021

Team Member

Ormond Beach, FL

- Worked register; prepared and assembled healthy foods and smoothies; helped to create a friendly work environment; regularly engaged with customers; handled customer complaints; helped with inventory; stocked shelves, refrigerator, and freezer; responsible for closing store (closed registers, food preparation for the next day, cleaned store)

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## VOLUNTEERING

### Libros de Familia

Aug. 2023 – May 2024

- Tutored an 8th-grade child who only knew Spanish in middle school Biology, Math, Reading, and Language Arts
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